

COONAWARRA, Australia

WMO: 948170

Lat: 37.2906S

Lon: 140.8253E

Elev: 189

StdP: 14.60

Time Zone: 9.50 (AUA)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	33.2	35.3	28.1	22.2	72.4	31.6	26.1	57.1	24.7	54.0	22.9	53.9	2.9	110	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.2	96.1	65.2	91.7	64.2	87.4	63.1	69.8	82.6	67.5	82.5	65.7	81.0	11.5	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	99.0	72.2	63.3	87.6	70.4	60.9	80.3	68.8	34.0	81.4	32.1	82.8	30.6	81.3	94.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	20.2	18.2	DB	29.0	106.0	1.9	3.9	27.6	108.8	26.5	111.1	25.4	113.2	24.0	116.1	(4)
(5)			WB	30.9	77.7	1.3	6.2	30.0	82.2	29.2	85.8	28.5	89.3	27.5	93.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.3	67.0	67.1	63.1	57.5	53.3	49.5	48.7	49.9	52.4	55.9	59.9	63.3	(6)	
	DBStd	8.86	7.63	7.43	7.19	6.01	5.00	3.96	3.71	4.13	4.92	6.40	7.36	7.43	(7)	
	HDD50	239	0	0	0	5	21	56	70	53	23	10	1	0	(8)	
	HDD65	3281	64	57	126	239	363	465	504	469	378	296	195	125	(9)	
	CDD50	2888	528	479	408	230	124	41	31	49	96	192	297	413	(10)	
	CDD65	457	128	116	69	14	2	0	0	0	1	13	41	73	(11)	
	CDH74	6547	1818	1466	950	204	10	0	0	0	15	219	666	1199	(12)	
	CDH80	3024	934	694	424	54	0	0	0	0	0	70	272	576	(13)	
(14) Wind	WSAvg	9.1	9.6	9.3	8.8	7.5	8.0	8.1	9.5	9.7	9.8	9.8	9.3	9.4	(14)	
(15) Precipitation	PrecAvg	25.3	1.2	0.9	1.2	1.7	2.3	3.3	3.6	3.5	2.8	1.9	1.5	1.4	(15)	
	PrecMax	34.5	4.1	2.2	3.2	3.6	5.4	6.9	5.6	6.4	5.2	3.9	3.7	4.6	(16)	
	PrecMin	16.7	0.1	0.0	0.2	0.2	0.3	0.6	1.7	1.0	1.2	0.1	0.5	0.2	(17)	
	PrecStd	4.2	1.1	0.6	0.7	0.9	1.2	1.4	1.1	1.5	1.1	1.1	0.7	1.1	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	103.6	98.9	97.0	86.6	74.5	64.5	62.1	69.0	76.2	87.8	93.1	99.7	(19)	
		MCWB	66.0	66.6	64.6	61.1	55.9	54.8	53.3	54.8	59.2	60.3	64.2	64.6	(20)	
	2%	DB	96.4	93.8	90.3	79.6	68.6	60.6	59.0	64.1	69.8	80.1	87.7	92.8	(21)	
		MCWB	65.8	65.2	63.1	58.9	55.6	53.3	52.0	53.2	56.4	59.1	61.6	64.0	(22)	
	5%	DB	90.4	88.8	84.5	74.5	64.9	58.6	57.2	60.6	65.8	74.1	82.1	86.4	(23)	
		MCWB	65.0	64.6	61.9	57.7	55.5	52.6	51.3	52.1	54.7	57.6	61.1	62.7	(24)	
	10%	DB	84.4	83.3	78.5	69.7	61.9	56.8	55.5	57.8	62.4	69.0	75.9	80.3	(25)	
		MCWB	63.2	63.6	61.2	56.9	54.7	51.7	50.5	50.9	53.7	56.3	60.1	61.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.2	72.4	69.4	63.9	60.4	58.5	56.0	57.3	61.3	67.6	69.9	71.8	(27)	
		MCDB	82.7	87.7	81.9	74.1	65.6	60.1	58.8	65.5	70.8	71.6	83.6	81.1	(28)	
	2%	WB	69.3	69.1	66.0	61.7	58.5	55.0	53.9	54.7	58.1	61.7	66.1	67.0	(29)	
		MCDB	84.0	83.1	81.1	71.4	63.5	58.5	57.1	60.6	66.5	72.0	80.0	80.8	(30)	
	5%	WB	67.1	66.8	64.3	60.0	57.1	53.6	52.5	53.1	56.2	59.3	63.6	64.6	(31)	
		MCDB	84.6	82.1	78.1	69.5	62.0	57.0	55.6	58.5	63.7	70.9	76.7	79.5	(32)	
	10%	WB	64.8	64.9	62.5	58.5	55.7	52.4	51.3	51.8	54.4	57.1	61.1	62.5	(33)	
		MCDB	81.4	80.0	75.8	66.9	60.6	55.8	54.4	56.7	61.0	67.3	73.5	78.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	27.2	25.8	24.7	22.6	16.8	15.0	13.8	15.5	17.8	21.2	23.6	26.2	(35)	
		MCDBR	37.6	34.7	34.3	30.9	20.6	17.0	14.4	19.2	23.2	30.4	34.1	38.0	(36)	
	5% WB	MCWBR	11.7	11.1	11.4	12.9	11.1	11.1	9.4	10.9	12.4	13.6	12.9	13.3	(37)	
		MCWBR	29.7	27.5	27.1	23.0	15.6	13.9	12.8	16.6	20.0	25.1	27.3	29.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.338	0.333	0.320	0.312	0.306	0.302	0.301	0.316	0.334	0.354	0.336	0.335	(40)	
		taud	2.499	2.524	2.558	2.565	2.564	2.562	2.557	2.500	2.441	2.381	2.471	2.495	(41)	
	Ebn at Noon		316	310	302	285	267	259	266	279	292	299	314	318	(42)	
		Edh at Noon	36	34	30	26	23	21	23	27	33	38	37	36	(43)	
(44) All-Sky Solar Radiation	RadAvg		2368	2059	1615	1120	746	609	658	934	1294	1726	2064	2309	(44)	
	RadStd		131	103	69	71	37	44	34	48	104	133	128	140	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (23 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	N/A	-173	+238	+70

Nomenclature: See separate page